

Reliability of Gaze Prediction for Stylized and Transformed Images

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Abstract

Recent advancements have shown increasingly accurate and fast visual gaze prediction models, which might now be feasible for real-time scenarios. These gaze prediction models would allow a program to infer where a user might focus on in an rendered scene, and might allow unique program behavior, rendering optimizations based on perception, or modelling of the user’s visual attention, without requiring direct eye-tracking hardware. Current state-of-the-art models use deep learning techniques, and most published image data sets for training are biased towards photography, at the expense of computer-rendered scenes, especially those with stylized subjects. We raise concerns over the generalizability of state-of-the-art models. We study a number of digital image transformations on existing datasets, using them as a reference point for what various rendering techniques or stylization effects might have on gaze prediction models. Our findings show that state-of-the-art models perform significantly worse on common image transformations. Our experiments reveal no significant, computable indicators of loss in accuracy, meaning more human gaze distribution data must be gathered in order to train and validate the generalizability of gaze prediction models.

CCS Concepts: • Computing Methodologies → Interest point and salient region detection; Cross-validation; Neural networks; Perception; Image processing; Non-photorealistic rendering.

Keywords: saliency, visual attention, generalizability, visual fixation prediction, rendering

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1 Introduction

Recent years have seen many advancements in the field of visual fixation prediction, which is the task of predicting the points at which a human will first focus on within the area of an image (sometimes called the “saliency” of an image). Droste et al. provide a new gaze prediction model, called UNISAL [4], which is among the top performing models on the MIT/Tuebingen Saliency Benchmark [7], and which generates predictions on frames in the DHF1K [11] dataset in nine milliseconds, moving gaze prediction tasks into the realm of real-time computing.

The most prominent published data sets for image saliency, including the MIT 300 [5] and CAT 2000 [1], provide only a handful of 3D computer-rendered scenes. Even the amount of stylized art, such as painting and collage, is noticeably lesser than that of photos of physical subjects in these datasets. The report from Kümmerer et al. [6] shows that leading gaze prediction models use deep-learning techniques, and the black-box nature of these techniques raises concerns that the models will not generalize well to all classes of images, especially those which are under-represented in the training data set.

2 Method

Although we wish to study model accuracy on computer-rendered images against benchmark results, we realize that it is difficult to isolate against the effects of uncontrolled variables when comparing two differing data sets. Instead, we hypothesize that digital transformations of a set of images, when compared to the untransformed images, will degrade the model’s predictive accuracy. We assume that by showing common digital transformations degrade accuracy, we show that the models cannot be expected to generalize well to many other domains of images, including stylized and computer-rendered images, especially those which have visual similarities to the transformations we test.

We utilize the dataset produced by Che et al. [3], which randomly selects 100 images from the CAT 2000 [1] dataset, produces corresponding transformed images for 18 transformation types, and measures gaze distributions of human subjects for all 1900 images. For each image, we measure its score on the NSS [10] and IG [8] metrics, and express those scores as percentages of the “gold standard” minus the “center bias” scores. The gold standard and the center bias are described by Kümmerer et al. [8] to be the “best guess” for an image with all image-specific human

gaze data considered, and with only dataset-specific human gaze data considered, respectively. Together, they provide a good upper bound and lower reference point with which to consider a model’s prediction accuracy. Expressing scores within this range allows us to find model degradation beyond that which might be expected due to a destructive or non-reversible transformation, as the degradation in performance produced by any destructive transformation should be captured by the gold standard and center bias for that transformation.

3 Results

We find that for both DeepGaze IIE [9] and UNISAL models, which hold top spots on the MIT/Tuebingen Saliency benchmark, all transformations except a simple horizontal mirror transformation result in loss of 5 to 95 percentage points of the center bias/gold standard range, for both IG and NSS metrics, relative to untransformed images. Untransformed images perform roughly in line with benchmark expectations, while almost all transformed images degrade to some degree beyond that which can be expected by loss of information during the transformation. These findings support our hypothesis, and seem to indicate that current models have poor generalizability to stylized images which bear resemblance to the common transformations we tested (including cropping, gaussian noise, contrast adjustment, rotation, shearing, motion blur, and edge enhancement).

3.1 Accuracy Loss Indicators

IG and NSS are the most useful metrics for measuring predictive accuracy, but we computed many others described by Bylinskii et al. [2] during our experiment. We were curious to see if any of the metrics we tested were indicators of loss in predictive accuracy for any transformation. Unfortunately, linear correlation revealed no strong trends between the change in any metric between untransformed and transformed images, and loss in accuracy. There is no method for revealing a model’s loss in predictive accuracy for a class of images other than measuring against human gaze distributions, according to the knowledge of the authors.

4 Conclusion

Our work indicates that state-of-the-art gaze prediction models cannot be relied on to produce predictions that are as accurate as benchmarks profess for images that are not well represented by prominent saliency data sets. This includes stylized and computer-rendered images. Unlocking gaze prediction for real-time rendering applications will likely require much greater amounts of training data and validation studies for these models.

Code for model inference, metric computation, and visualization of data; as well as plaintext data for our results; can be found at our repository [12].

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